



SIMATS ENGINEERING
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CSA1402-COMPILER DESIGN

Compiler Construction Tools: LEX And YACC

Automate Compiler Development, Saving Time And Effort.

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Introduction to Compiler Construction Tools



- LEX and YACC are essential tools that streamline the development of compilers and interpreters.
- They automate the process of lexical analysis and syntax parsing, reducing manual coding efforts.
- Using these tools helps in creating more reliable and maintainable compiler components efficiently.

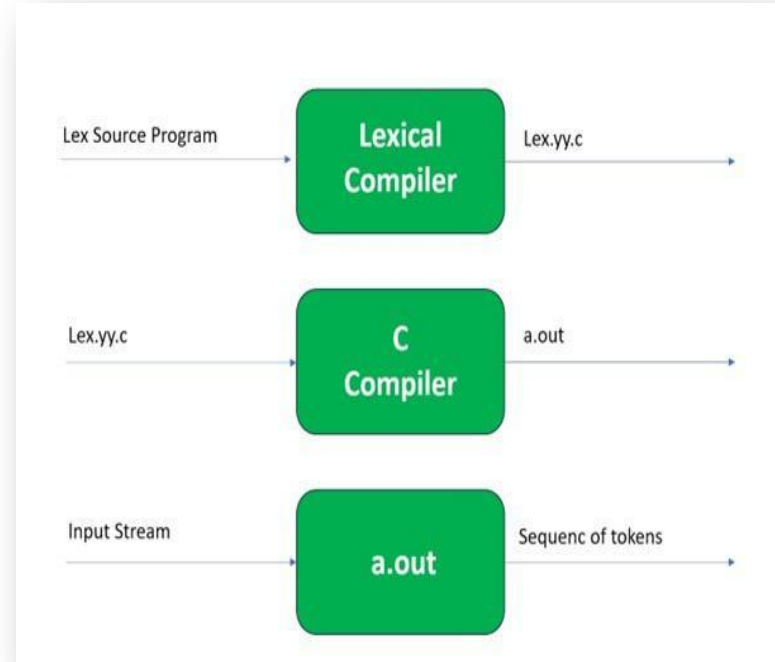


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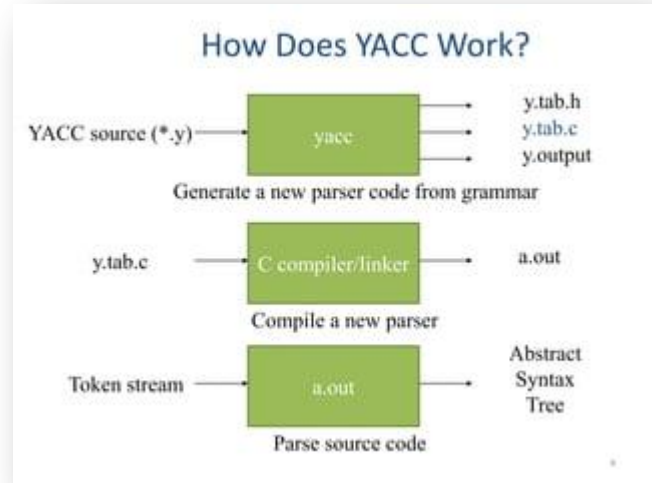
Overview of LEX

- LEX is a lexical analyzer generator that converts regular expressions into C code for token recognition.
- It simplifies the process of defining tokens and their patterns in a concise, declarative manner.
- LEX outputs a scanner that can be integrated into a compiler to efficiently perform lexical analysis.



Overview of YACC

- YACC (Yet Another Compiler Compiler) generates a parser based on a formal grammar description.
- It automates the syntax analysis phase by producing C code that implements the parser logic.
- YACC supports complex language grammars, including ambiguous and recursive structures, with minimal manual effort.



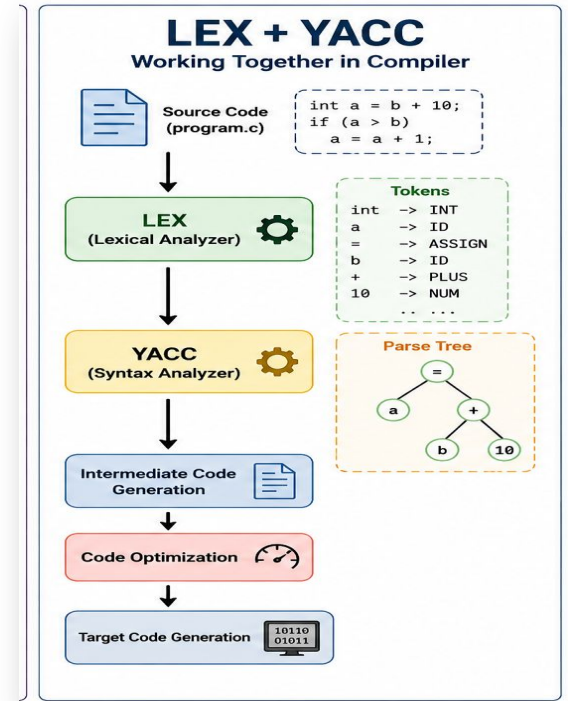


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How LEX and YACC Work Together



- LEX handles tokenization, breaking source code into manageable symbols for the parser.
- YACC takes these tokens and constructs a syntax tree based on the language grammar.
- Integrating LEX and YACC creates a powerful pipeline that automates major parts of compiler construction.





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Benefits of Using LEX and YACC



- They significantly reduce development time by automating routine and error-prone tasks.
- These tools improve the accuracy and consistency of the compiler components.
- They facilitate easier maintenance and modifications of the compiler as language specifications evolve.

Conclusion and Impact



- LEX and YACC are foundational tools that have transformed compiler construction practices.
- Their automation capabilities allow developers to focus on language semantics and optimization.
- Overall, they contribute to faster, more reliable compiler development with less manual effort.

THANK YOU

